Karthikeya Kumar

Software Engineer — Backend Engineer

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Skills

Programming: Python, Go, C++, SQL

Backend: FastAPI, REST APIs, Celery, Redis, Kafka, WebSockets

Databases: PostgreSQL, MongoDB, ChromaDB, Neo4j

Systems & Tools: Docker, Git, GitHub, Linux, AWS

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

AI/ML: TensorFlow, PyTorch, Scikit-learn, RAG, LangChain

Experience

Data Analysis Intern Sep 2025 – Oct 2025

SkillFied Mentor (Remote)

- Designed automated data validation workflows using SQL and Python to process datasets containing **10,000+ records**.
- Built automated deduplication pipelines that identified and flagged duplicate records with **99% accuracy**.
- Streamlined data processing and reporting workflows by eliminating inconsistencies and improving dataset quality.

Projects

AI Co-Founder — Multi-Agent AI Business Platform GitHub

- Architected a FastAPI backend with **10 specialized agents**, effort-based routing, Redis-backed resource budgets, and per-(company, effort) agent caching.
- Engineered a hybrid **RAG pipeline** using semantic + keyword retrieval with pgvector HNSW indexes, achieving **95% pass rate**, **0.950 Recall@5**, and **0.883 MRR**.
- Built an **Apache Kafka** event-driven pipeline for message persistence, chat-memory extraction, and session-title generation with at-least-once delivery and isolated consumer failures.
- Implemented multi-tier **Redis caching** for embeddings, sessions, company data, and agent state, reducing worst-case latency from **15s to 2s** and repeat reads to near-zero latency.
- Added **WebSocket observability** for real-time tool execution traces, sub-agent lifecycle events, heartbeats, and per-session error handling.

TripSync AI — Travel Recommendation Platform GitHub Live

Python, FastAPI, PostgreSQL, React, Vite, TensorFlow, Docker

- Developed a FastAPI backend exposing **5 REST APIs** for recommendations, destination lookup, itinerary generation, hotel ranking, and health checks.
- Designed PostgreSQL ingestion pipelines for **75-column destination** and **28-column hotel** datasets using JSON/CSV preprocessing and validation.
- Built a React trip-planning workflow with request validation and deterministic response shaping for recommendation and itinerary APIs.

OmniMark — Automated Answer Evaluation Platform GitHub

Python, FastAPI, Celery, React, Docker, MongoDB, AWS

- Architected a Dockerized **3-service platform** with FastAPI, Celery, React/Nginx, JWT authentication, **15+ REST APIs**, MongoDB, and AWS EC2 deployment.
- Built asynchronous Celery pipelines for OCR, grading, plagiarism detection, and real-time progress tracking of PDF processing workflows.
- Implemented a plagiarism engine performing **5,000+ pairwise comparisons** and used Strategy/Factory patterns for modular grading, auditing, analytics, and exports.

Education

Aditya University 2024 – 2028

B.Tech in Artificial Intelligence and Machine Learning CGPA: 7.73/10

Achievements

- Secured **1st Prize** at TechnicalHub Project Space 8.0 for developing an automated paper evaluation platform.
- Won **1st Prize** in the TechnicalHub Power BI Competition.

Relevant Coursework

Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering